The Impact of Music Packaging on Listener Utility and Simulated Music Popularity

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1. Introduction

Music packaging is widely used in the music industry to attract listeners. However, whether packaging itself can change listener choice remains unclear. Packaging plays an important role not only in music industry, it also includes branding, social media, and product marketing. This project uses simulated model to investigate the effect of music packaging on listener choice.

2. Building Model

We assume diminishing marginal benefits of packaging intensity:

$$\boldsymbol{U = \log(P + 1) + \varepsilon}$$

U is the utility function of each listener with 1 observable explanatory variable P, and one stochastic error term ε. P stands for the intense of the packaging, P $\in$ [0, ∞); $\varepsilon \sim$ N (0, 1), which included social influence, personal tastes, quality and other related factors.

$$\frac{dU}{dP} = \frac{1}{(P+1)} > 0$$

$$\frac{d^2U}{dP^2} = -\frac{1}{(P+1)^2} < 0$$

Therefore, the utility function is strictly concave and exhibits a diminishing marginal benefit of packaging intensity.

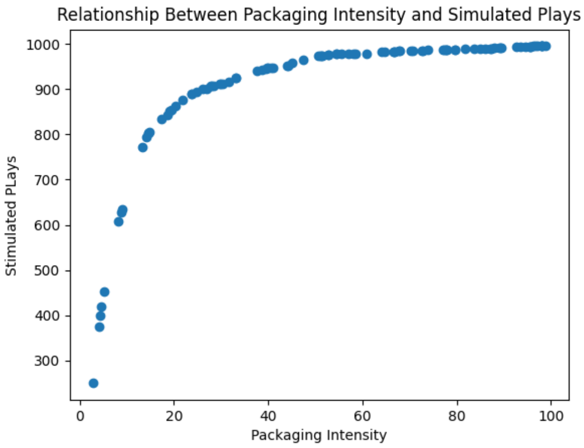
3. Methodology

The simulation generates 100 songs with uniformly distributed packaging intensity with range [0,100) and 1000 listener utility function, which $\varepsilon \sim$ N (0, 1). Each song was evaluated by one listener, repeat this process until 1000 listeners finished the evaluation. If the utility is greater than 2, listeners will choose to play the song, which reflected on scatter plot.

To further examine the relationship between packaging and utility, an additional analysis was conducted. Instead of simulated listener choices and play counts, I focused on plotting graphs of packaging intensity (P) against utility function (U); two scenarios were considered: $\varepsilon \sim$ N (0, 1) and $\varepsilon \sim$ N (0, 0.1^2), to illustrate the effect of random variation on the observed relationship.

4. Results

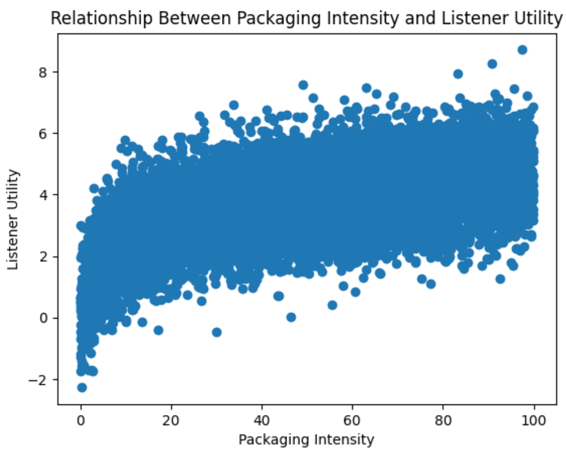
Figure 1 shows the relationship between packaging intensity and simulated plays. The results indicate that the marginal impact of packaging diminishes as packaging intensity increases. At low levels of packaging intensity, simulated plays increase rapidly. However, the growth slows as packaging intensity becomes stronger.



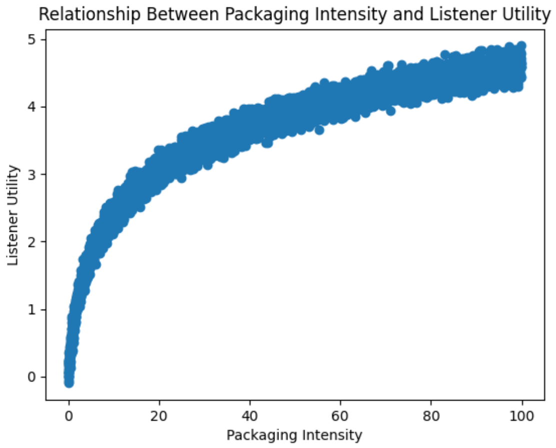
(Figure 1)

5. Additional Analysis

Additional analysis will focus on the relationship between music packaging and listener utility. In order to clearly illustrate the relationship, the large amount variation in Figure 2 obscures the relationship between packaging and utility, which has one standard deviation. Figure 3 shows the same relationship with 0.1 standard deviation. Therefore, due to smaller standard deviation, the relationship between packaging and utility becomes more apparent in Figure 3.



(Figure 2)



(Figure 3)

6. Results for Additional Analysis

According to Figure 3, the curve is similar to Figure 1, marginal benefit diminishes as packaging intensity increases, which can also illustrate the results in simulation model. However, the error term in real world cannot be this small, as Figure 2 shown, if we only focus on packaging, the random shock could disperse the impact.

7. Discussion

The results illustrated packaging appears to be more influential when initial packaging intensity is low. For music producer, packaging may be useful for attracting listeners, but it cannot guarantee unlimited growth. The impact of packaging may vary depending on external factors. Music producers should not only focus on packaging, but also consider external effects.

8. Limitations

The model of packaging and popularity does have many limitations, the impact of packaging is not absolutely independent; music quality, public tastes, artists reputation, and social influence could affect the results. According to additional analysis, with a large random shock, the impact of packaging will be dispersed. Therefore, this model is only adapted to a certain condition world, as other factors have little affect and independent to packaging.

Another limitation of this project is that the relationship between packaging intensity and listener choice are already determined by utility function. The logarithm in utility function described diminishing marginal benefits of packaging. However, different utility function may have different results, for instance linear or quadratic functions. Therefore, the results should be interpreted as a prediction in a specific situation rather than real-world phenomenon.

9. Conclusion

This project investigated the relationship between packaging and listener choice through a simplified simulated model. Additional analysis also suggested that the impact of packaging is sensitive to random shock.

The results suggested stronger packaging intensity increase simulated plays. However, the marginal effect diminishes as increases packaging intensity. For music producers, packaging might be useful to attract listeners, but not forever because they also have to consider other factors.